Table of Contents

Introduction

As part of MATH1312 Assignment 2, this report examines two different datasets using multiple linear regression methods.

Question 1 examines a pharmacological investigation on rats. The goal is to calculate the proportion of drug dose retained in the liver (Y) using three predictors: body weight (X_1), liver weight (X_2), and dose administered (X_3). The study involves fitting a full regression model, determining variable significance, checking for multicollinearity, and using backward elimination to enhance the model.

Question 2 examines Crest toothpaste sales data to identify significant marketing and sales aspects that influence consumer sales. This includes determining overall model relevance, assessing individual predictors, verifying regression assumptions (linearity, normality, homoscedasticity, and independence), and comparing various model selection strategies such as forward, backward, and stepwise regression.

The study uses a variety of statistical tools, including ANOVA, t-tests, correlation analysis, variance inflation factors (VIF), and residual diagnostics, to assess the strength, reliability, and interpretability of both regression models.

Methodology

The analysis was carried out using the statistical software R, applying multiple linear regression techniques to both datasets provided in the assignment.

Question 1 – Liver Data Analysis

The first dataset focused on estimating the percentage of drugs dose retained in the liver (Y) of rats using three predictors:

X_1: Body weight of the rat

X_2: Liver weight

X_3: Dose given

All three predictors were used to create a comprehensive multiple linear regression model. To validate the linear algebra, we computed the inverse of the matrix X, $X'X$.

The model was then assessed using:

- T-tests to determine the individual importance of predictors
- ANOVA (F-test) to determine overall model significance

To discover multicollinearity, we used the correlation matrix and variance inflation factors (VIF).

To simplify the model by removing insignificant variables, backward elimination was used with a Fstay = 4.

Question 2 – Crest Toothpaste Sales Data

The second dataset investigated at how marketing and sales characteristics influenced toothpaste sales. A multiple linear regression model was first fitted with all relevant predictors. The following measures were taken:

- The overall model significance was examined using the ANOVA table.
- Individual predictor significance was assessed using t-tests.

Residual diagnostics were conducted to verify linear regression assumptions:

- Linearity
- Normality (Shapiro-Wilk test)
- Homoscedasticity (non-constant variance test)
- Independence (Durbin-Watson test)
- Multicollinearity was examined using the VIF and correlation matrix

Model selection techniques were applied, including:

- Forward selection
- Backward elimination
- Stepwise regression

All results were interpreted in context, and the final models were selected based on their simplicity, statistical significance, and adherence to regression assumptions.

Result

Question 1

Part A

Design Matrix X

```
> X
      [,1] [,2] [,3] [,4]
 [1,]    1  176  6.5 0.88
 [2,]    1  176  9.5 0.88
 [3,]    1  190  9.0 1.00
 [4,]    1  176  8.9 0.88
 [5,]    1  200  7.2 1.00
 [6,]    1  167  8.9 0.83
 [7,]    1  188  8.0 0.94
 [8,]    1  195 10.0 0.98
 [9,]    1  176  8.0 0.88
[10,]    1  165  7.9 0.84
[11,]    1  158  6.9 0.80
[12,]    1  148  7.3 0.74
[13,]    1  149  5.2 0.75
[14,]    1  163  8.4 0.81
[15,]    1  170  7.2 0.85
[16,]    1  186  6.8 0.94
[17,]    1  146  7.3 0.73
[18,]    1  181  9.0 0.90
[19,]    1  149  6.4 0.75
```

Figure 1: Design Matrix

Transpose of X

```
> x_transpose
       [,1]   [,2] [,3]   [,4]  [,5]   [,6]   [,7]   [,8]   [,9]  [,10] [,11]  [,12]  [,13]  [,14]  [,15]  [,16]  [,17] [,18]  [,19]
[1,]   1.00   1.00    1   1.00   1.0   1.00   1.00   1.00   1.00   1.00   1.0   1.00   1.00   1.00   1.00   1.00   1.00   1.0   1.00
[2,] 176.00 176.00  190 176.00 200.0 167.00 188.00 195.00 176.00 165.00 158.0 148.00 149.00 163.00 170.00 186.00 146.00 181.0 149.00
[3,]   6.50   9.50    9   8.90   7.2   8.90   8.00  10.00   8.00   7.90   6.9   7.30   5.20   8.40   7.20   6.80   7.30   9.0   6.40
[4,]   0.88   0.88    1   0.88   1.0   0.83   0.94   0.98   0.88   0.84   0.8   0.74   0.75   0.81   0.85   0.94   0.73   0.9   0.75
```

Figure 2: Transpose Matrix

Matrix (X′X)−1

```
> XtX_inverse
            [,1]         [,2]         [,3]       [,4]
[1,]  6.33809378 -0.074426957 -0.068005387   8.133435
[2,] -0.07442696  0.010644476 -0.002783671  -2.006297
[3,] -0.06800539 -0.002783671  0.049620475   0.183175
[4,]  8.13343533 -2.006296702  0.183175008 388.083181
```

Figure 3 :Matrix (X'X)−1

Part B

The full multiple linear regression model was fitted using all three predictors:

$$Y = \beta 0 + \beta 1X1 + \beta 2X2 + \beta 3X3 + \varepsilon$$

Here:

X1 = body weight of rat

X2 = weight of liver

X3 = dose given

Y = percentage of dose in liver.

```
Call:
lm(formula = Y ~ X1 + X2 + X3, data = liver)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.100557 -0.063233  0.007131  0.045971  0.134691 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)  
(Intercept)  0.265922   0.194585   1.367   0.1919  
X1          -0.021246   0.007974  -2.664   0.0177 *
X2           0.014298   0.017217   0.830   0.4193  
X3           4.178111   1.522625   2.744   0.0151 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07729 on 15 degrees of freedom
Multiple R-squared:  0.3639,    Adjusted R-squared:  0.2367 
F-statistic:  2.86 on 3 and 15 DF,  p-value: 0.07197
```

Figure 4: Model Summary

From the R output we find

- Intercept = 0.2659
- X1 (Body Weight) = -0.0212
- X2 (Liver Weight) = 0.0143
- X3 (Dose Given) = 4.1781

The fitted regression equation is:

$$Y = 0.2659 - 0.0212 * X1 + 0.0143 * X2 + 4.1781 * X3 + \varepsilon$$

Inference: The coefficient for body weight (X1) is statistically significant ($p = 0.0177$), showing that for every 1 gram increase in body weight, the percentage of dose reduces by roughly 2.12%, when all other factors remain unchanged. Dose (X3) is also significant ($p = 0.0151$), indicating that a 1 unit increase in dose results in a 4.18% increase in retention. However, liver weight (X2) is not statistically significant ($p = 0.4193$) and may not make a substantial contribution to the model. The model explains 36.4% of the variation in Y ($R^2 = 0.3639$). However, the adjusted R^2 decreases to 0.2367, indicating weak predictive power. The overall F-test p-value is 0.07197, indicating that the model is non-significant at the 5% level but marginally significant at the 10% level. This means that there is moderate evidence that the factors explain variation in liver retention.

Part C

```
Analysis of Variance Table

Response: Y
          Df   Sum Sq  Mean Sq F value  Pr(>F)
X1         1 0.003216 0.003216  0.5383 0.47446
X2         1 0.003067 0.003067  0.5134 0.48467
X3         1 0.044982 0.044982  7.5296 0.01507 *
Residuals 15 0.089609 0.005974
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 5: ANOVA table of the Model

ANOVA Hypothesis

H0: $\beta_i = 0$, a linear relationship **does not** exist

Ha: $\beta_i \neq 0$, a linear relationship **does** exist

df = 3, 15

F-critical = 3.287 (from F-table, at $\alpha = 0.05$)

Inference: The ANOVA table shows partial F-test results for each predictor variable in the multiple regression model. These tests determine if each predictor contributes significantly to explaining variation in the response variable while controlling for other variables. We compare each F-value to the F-critical value of 3.287 at the 5% level of significance.

- X1 (Body Weight) has an F-value of 0.5383, which is lower than F-critical (3.287). This shows that X1 is not statistically significant, and we fail to reject the null hypothesis for this variable.
- X2 (Liver Weight) has an F-value of 0.5134, which is similarly below the critical threshold and hence not significant.
- X3 (Dose) has an F-value of 7.5296, which is greater than F-critical (3.287), indicating that we reject the null hypothesis and conclude that dose is a significant predictor.

Based on this comparison, X3 (Dose) should remain in the model, whereas X1 (Body Weight) and X2 (Liver Weight) can be eliminated because their F-values do not exceed the critical value.

```
Call:
lm(formula = Y ~ X1 + X2 + X3, data = liver)

Residuals:
      Min        1Q    Median        3Q       Max
-0.100557 -0.063233  0.007131  0.045971  0.134691

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.265922   0.194585   1.367   0.1919
X1          -0.021246   0.007974  -2.664   0.0177 *
X2           0.014298   0.017217   0.830   0.4193
X3           4.178111   1.522625   2.744   0.0151 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07729 on 15 degrees of freedom
Multiple R-squared:  0.3639,    Adjusted R-squared:  0.2367
F-statistic:  2.86 on 3 and 15 DF,  p-value: 0.07197
```

Figure 6: Summary of the Model for t-test

Hypothesis for Each Coefficient

H0: βj = 0, the predictor(s) is statistically insignificant

Ha: βj != 0, the predictor(s) is statistically significant

df = 15

t-critical = 2.131 (from t-table, at α =0.05)

Inference: A two-tailed t-test with α = 0.05 was used to assess the significance of each regression coefficient. The t-critical value at 15 degrees of freedom was calculated to be ±2.131. Each coefficient's t-value was compared to this critical value.

- The t-value for X1 (Body Weight) is -2.664, which is smaller than -2.131, leading us to reject the null hypothesis and conclude that X1 is a statistically significant predictor.
- The t-value for X2 (Liver Weight) is 0.830, which is within the range of -2.131 to 2.131. As a result, we fail to reject the null hypothesis, and X2 is not statistically significant.
- The t-value for X3 (Dose) is 2.744, which is larger than 2.131, leading us to reject the null hypothesis and conclude that X3 is a significant predictor.

Based on t-test, X1 and X3 are significant and should remain in the model, whereas X2 can be removed.

Conclusion

Based on the results of both the ANOVA partial F-tests and the individual t-tests, we can conclude the following:

- X3 (Dose) is consistently determined to be statistically significant across both tests. The F-value (7.5296) and t-value (2.744) indicate a significant effect on the response variable, exceeding the critical values (3.287 and ±2.131, respectively). It should remain in the model.
- X1 (Body Weight) yields inconsistent results: it is not significant in the ANOVA test (F = 0.5383 < 3.287), but significant in the t-test (t = -2.664 < -2.131). This implies that, while its unique contribution in the presence of other variables may be minimal, it still has a meaningful association with the response and should be evaluated for retention.
- In both tests, X2 (Liver Weight) had no significant effect on the response (F = 0.5134 < 3.287; t = 0.830 within ±2.131). Therefore, X2 can be removed from the model.

In conclusion, X3 (Dose) certainly ought to be kept, X2 (Liver Weight) should be excluded, and X1 (Body Weight) may be retained based on its t-test result, particularly if model selection criteria or diagnostics support inclusion.

Part D

```
> # Variance Inflation Factor (VIF)
> vif(model)
       X1        X2        X3 
52.101911  1.335679 51.427147 
```

Figure 7: VIF values.

A VIF value greater than 10 usually indicates problematic multicollinearity.

Inference: The VIF for X1 (Body Weight) and X3 (Dose) were 52.10 and 51.43, respectively, indicating severe multicollinearity. In contrast, X2 (Liver Weight) had a VIF of 1.34, demonstrating the absence of multicollinearity. According to these findings, there is substantial evidence of multicollinearity between X1 and X3, which may impair the stability of the coefficient estimates and should be addressed during model development.

Part E

```
> # Backward elimination with Fstay = 4
> model_back <- step(model, direction = "backward", k = 4)
Start:  AIC=-85.78
Y ~ X1 + X2 + X3

       Df Sum of Sq      RSS     AIC
- X2    1  0.004120 0.093729 -88.924
<none>              0.089609 -85.778
- X1    1  0.042408 0.132017 -82.416
- X3    1  0.044982 0.134591 -82.049

Step:  AIC=-88.92
Y ~ X1 + X3

       Df Sum of Sq      RSS     AIC
<none>              0.093729 -88.924
- X1    1  0.039851 0.133580 -86.192
- X3    1  0.043929 0.137658 -85.621
```

Figure 8: Backward Elimination.

```
> summary(model_back)

Call:
lm(formula = Y ~ X1 + X3, data = liver)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.12333 -0.07416  0.01238  0.04884  0.12668 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)  
(Intercept)  0.285517   0.191267   1.493   0.1550  
X1          -0.020444   0.007838  -2.608   0.0190 *
X3           4.125329   1.506472   2.738   0.0146 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07654 on 16 degrees of freedom
Multiple R-squared:  0.3347,	Adjusted R-squared:  0.2515 
F-statistic: 4.024 on 2 and 16 DF,  p-value: 0.0384
```

Figure 9: New Model Summary.

To simplify the model, backward elimination was performed starting with all three predictors: X1 (Body Weight), X2 (Liver Weight), and X3 (Dose). The elimination was based on given Fstay = 4.

Inference: The method began with the entire model, and at each stage, the predictor with the lowest contribution (based on AIC and F-value) was considered for elimination. The variable X2 (Liver Weight) was removed first because it made the smallest impact, and its elimination improved the model's AIC from -85.78 to -88.92.

The final, more parsimonious model included X1 (Body Weight) and X3 (Dose). The remaining predictors were both statistically significant (p-values of 0.0190 and 0.0146, respectively). The fitted model was:

$$\mathbf{Y = 0.2855 - 0.0204*X1 + 4.1253*X3 + \varepsilon}$$

The model explains 33.5% of the variability in the percentage of dosage retained, ($R^2 = 0.3347$) and is statistically significant (F-statistic = 4.024, p-value = 0.0384). As a result, the final reduced model includes X1 and X3 as significant predictors while excluding X2, yielding a more efficient and interpretable model without losing explanatory power.

Question 2

<u>Part A</u>

Scatter Plot of the data

The ggpairs() function was used to generate a scatterplot matrix that visually examined the correlations between the response variable (Crest sales) and each of the predictor factors (x1, x2, x3).

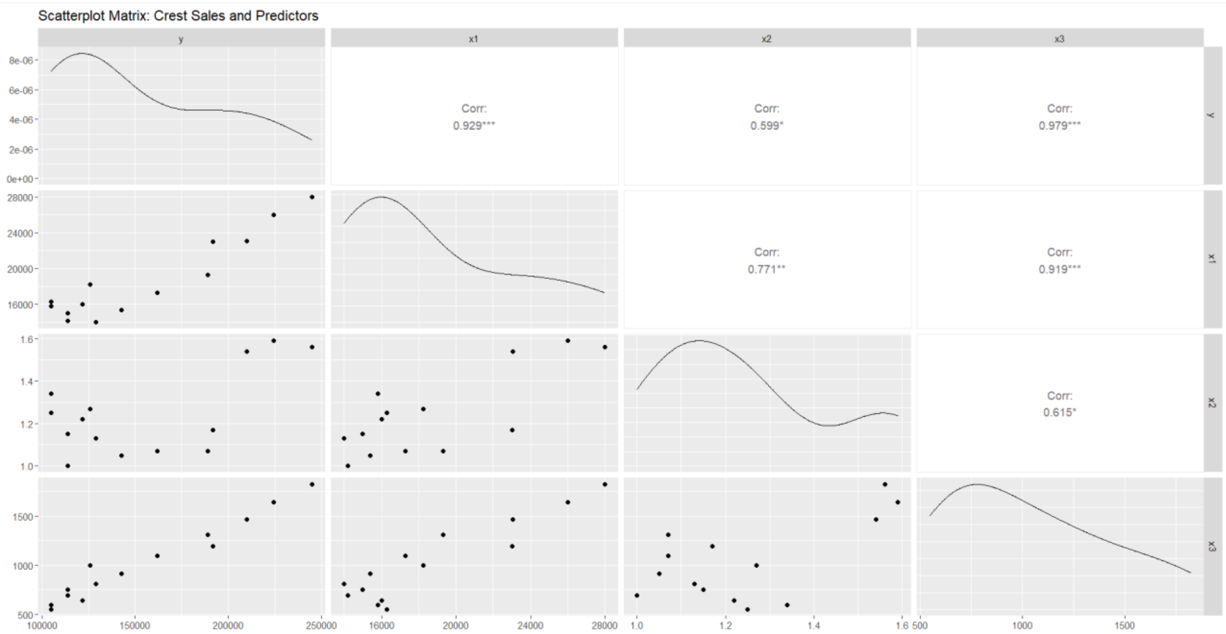

Figure 10: Scatter Plot of Crest Data.

Inference:

Sales vs. Budget (y vs. x1)

Sales and Budget (y vs. x1) have a strong positive linear correlation = 0.929. The scatterplot verifies this with an increasing trend, implying that larger advertising expenditure is strongly linked to higher Crest sales.

Sales vs. Income (y vs. x3)

The correlation is significantly larger, at 0.979, indicating a very strong and nearly perfect correlation. As disposable income improves, Crest toothpaste sales rise dramatically, most likely due to increased customer purchasing power.

Sales vs. Ratio (y vs. x2)

The correlation is 0.599, which indicates a slight positive relationship. The scatterplot indicates greater variability and a less distinct pattern, implying that the ratio may not be as effective a predictor as x1 or x3.

Correlations among Predictors:

- x1 and x3: Correlation = 0.919, very strong multicollinearity.
- x1 and x2: Correlation = 0.771, moderately strong
- x2 and x3: Correlation = 0.615, moderate

Based on the scatter plots and correlation values, x1 (budget) and x3 (income) have strong linear correlations with Crest sales and are expected to be the most important factors in the regression model. However, the significant correlation between x1 and x3 suggests the possibility of multicollinearity, which should be confirmed by VIF values before finishing the model.

Model Summary

```
> summary(model_crest)

Call:
lm(formula = y ~ x1 + x2 + x3, data = crest)

Residuals:
   Min     1Q Median     3Q    Max 
-24088  -2568   1021   3836  10100 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  34104.559  17654.144   1.932 0.082187 .  
x1               3.746      1.976   1.896 0.087243 .  
x2          -30046.343  22859.674  -1.314 0.218066    
x3              85.926     17.911   4.797 0.000727 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9574 on 10 degrees of freedom
Multiple R-squared:  0.9691,	Adjusted R-squared:  0.9598 
F-statistic: 104.5 on 3 and 10 DF,  p-value: 7.537e-08
```

Figure 11: Model Summary of Crest Toothpaste data.

From the R output we find:

- Intercept = 34,104.56
- x1 (Advertising Budget) = 3.75
- x2 (Sales Ratio) = −30,046.34
- x3 (Disposable Income) = 85.93

The fitted regression equation is:

$$\mathbf{y = 34104.56 + 3.75*x1 - 30046.34*x2 + 85.93*x3 + \varepsilon}$$

Inference: To predict Crest sales (y), a multiple linear regression model was developed with predictors x1 (advertising budget), x2 (ratio), and x3 (US personal disposable income).

The model shows an excellent fit, with a Multiple R-squared of 0.9691 and an Adjusted R-squared of 0.9598, indicating that the three predictors account for roughly 96% of Crest sales variance. The overall model is highly significant, with an F-statistic of 104.5 and a p-value of 7.54×10^{-8}.

In terms of individual predictors:

- Income (x3) has a significant ($p < 0.001$) positive impact on Crest sales.
- x1 (budget) has a marginal significance level ($p = 0.087$), indicating that it may make a meaningful contribution but is on the borderline.
- x2 (ratio) is not statistically significant ($p = 0.218$), implying that it may not be an effective predictor in the context of x1 and x3.

These results emphasize the significant impact of consumer income (x3) and advertising budget (x1) on sales, although the sales ratio (x2) may be less significant.

ANOVA table for predictor significance

```
> anova(model_crest)
Analysis of Variance Table

Response: y
          Df     Sum Sq    Mean Sq F value    Pr(>F)    
x1         1 2.5611e+10 2.5611e+10 279.392  1.23e-08 ***
x2         1 1.0202e+09 1.0202e+09  11.130 0.0075401 ** 
x3         1 2.1097e+09 2.1097e+09  23.015 0.0007265 ***
Residuals 10 9.1667e+08 9.1667e+07                      
```

Figure 12: Anova Table of crest data.

ANOVA Hypothesis

H_0: $\beta_i = 0$, The predictor has no significant contribution to Crest sales

H_a: $\beta_i \neq 0$, The predictor significantly contributes to Crest sales

Inference: The ANOVA table shows partial F-test results for each predictor variable in the multiple regression model. The p-values are compared to the significance level ($\alpha = 0.05$) to evaluate if each predictor significantly contributes to the variation in Crest sales.

x1 (Advertising Budget):

F-value = 279.39

p-value = 1.23×10^{-8}

Since the p-value is less than 0.05, hence we reject H_0 and conclude that x1 is a highly significant predictor of Crest sales.

x2 (Sales Ratio):

F-value = 11.13

p-value = 0.00754

Since the p-value is less than 0.05, hence we reject H0 and conclude that x2 is a highly significant predictor of Crest sales.

x3 (Disposable Income):

F-value = 23.02

p-value = 0.00073

Since the p-value is less than 0.05, hence we reject H0 and conclude that x3 is a highly significant predictor of Crest sales.

At the 5% significance level, all three predictors (x1, x2, and x3) are statistically significant. Each contributes value to the model, justifying their inclusion in the regression analysis for predicting Crest sales.

Part B

t-test for coefficient significance

```
> summary(model_crest)

Call:
lm(formula = y ~ x1 + x2 + x3, data = crest)

Residuals:
   Min     1Q Median     3Q    Max 
-24088  -2568   1021   3836  10100 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  34104.559  17654.144   1.932 0.082187 .  
x1               3.746      1.976   1.896 0.087243 .  
x2          -30046.343  22859.674  -1.314 0.218066    
x3              85.926     17.911   4.797 0.000727 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9574 on 10 degrees of freedom
Multiple R-squared:  0.9691,	Adjusted R-squared:  0.9598 
F-statistic: 104.5 on 3 and 10 DF,  p-value: 7.537e-08
```

Figure 13: t-test.

Hypotheses for each coefficient β_j:

Null Hypothesis (H0): $\beta_j = 0$, the predictor is not statistically significant (no linear effect on sales)

Alternative Hypothesis (H_a): $\beta_j \neq 0$, the predictor is statistically significant (has a linear effect on sales)

Significance Level (α): 0.05

Inference: Each t-test determines whether a certain coefficient is significantly different from zero. We compare the p-values with $\alpha = 0.05$.

x1 (Advertising Budget):

t = 1.896

p-value = 0.0872

Since the p-value is greater than 0.05, hence we fail to reject H_0 and conclude that x1 is not significant predictor of Crest sales.

x2 (Sales Ratio):

t = −1.314

p-value = 0.2181

Since the p-value is greater than 0.05, hence we fail to reject H_0 and conclude that x2 is not significant predictor of Crest sales.

x3 (Disposable Income):

t = 4.797

p-value = 0.0007

Since the p-value is less than 0.05, hence we reject H_0 and conclude that x2 is a significant predictor of Crest sales.

The t-tests show that only x3 (disposable income) is statistically significant at the 5% level. Although x1 (budget) is not statistically significant at $\alpha = 0.05$, it may be considered moderately significant at 10% level. x2 (sales ratio) is not significant and can be eliminated in a simpler model.

Confidence Intervals (95%)

The 95% confidence intervals provide a range of possible values for each regression coefficient. A confidence interval that includes zero indicates that the coefficient may not be substantially different from zero at the 5% significance level.

```
> # Confidence Intervals (95%)
> confint(model_crest)
                    2.5 %       97.5 %
(Intercept) -5.231326e+03 73440.444468
x1          -6.569538e-01     8.148674
x2          -8.098087e+04 20888.185496
x3           4.601772e+01   125.835218
```

Figure 14: Confidence Intervals for coefficients and intercept.

Intercept:

95% CI: (−5,231.33, 73,440.44)

The interval is wide and includes 0.

Inference: The intercept is not statistically significant.

x1 (Advertising Budget):

95% CI: (−0.657, 8.15)

The interval includes 0.

Inference: x1 is not significant at the 5% level, consistent with the t-test result.

x2 (Sales Ratio):

95% CI: (−80,980.87, 20,888.19)

The interval is wide and includes 0.

Inference: x2 is not statistically significant.

x3 (Disposable Income):

95% CI: (46.02, 125.84)

The entire interval is above zero.

Inference: x3 is a statistically significant predictor of Crest sales. We are 95% confident that for every unit increase in disposable income, sales increase by between 46 and 126 units.

Only x3 (disposable income) has a 95% confidence interval that excludes zero, indicating that it is a statistically significant predictor of Crest sales. The other predictors (x1 and x2) have intervals that contain zero, which is consistent with the t-test results and indicates lesser predictive reliability.

<u>Part C & D</u>

Residual Analysis

To ensure that the regression model is valid, and the statistical inferences are credible, the following assumptions must be satisfied:

1. **Linearity**
   The relationship between the independent and dependent variables is linear.
2. **Independence**
   The residuals (errors) are independent. This is particularly important when working with time series data.
3. **Homoscedasticity (Constant Variance)**
   The residuals have constant variance at all levels of the independent variables.
4. **Normality of Residuals**
   The residuals are normally distributed.
5. **No Multicollinearity**
   Independent variables are not highly correlated with each other.

The residual diagnostic plots are used to determine whether the regression model meets the following assumptions: linearity, normality of errors, homoscedasticity (constant variance), and observation independence.

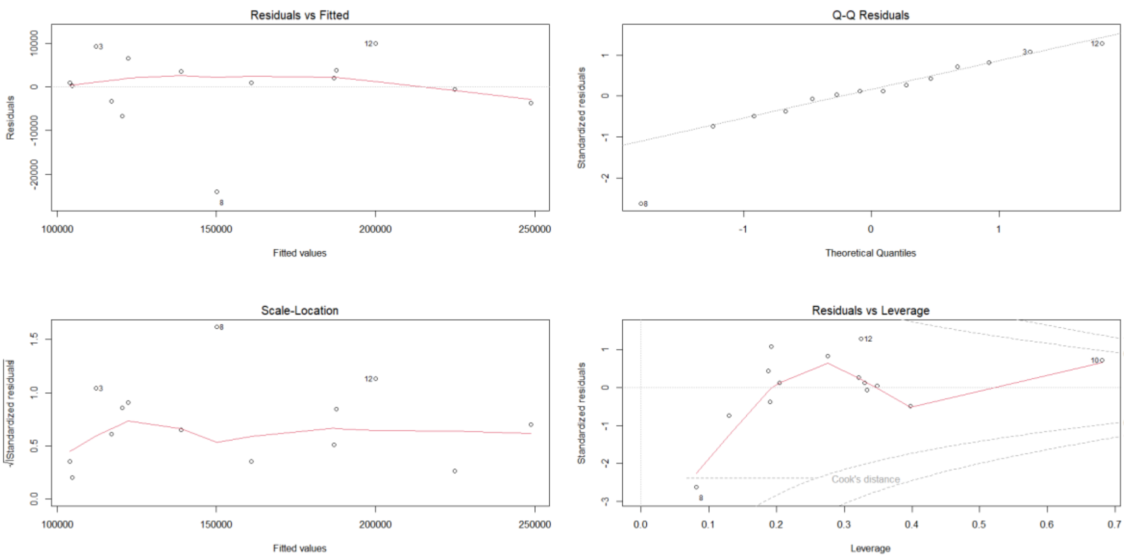

Figure 15: Residual Plots.

Inference:

Residuals vs Fitted (Top-Left Plot)

This plot assesses linearity and homoscedasticity. The residuals are mainly randomly distributed around zero, with some modest curvature and probable outliers (particularly observation 8). There is no significant violation of linearity, but some moderate nonlinearity or heteroscedasticity may exist.

Normal Q-Q Plot (Top-Right Plot)

This graph evaluates whether the residuals are normally distributed. Except for minor deviations at the tails (observation 8), most points follow the q-q line. The residuals are approximately normal, with just slight deviations at the extremes.

Scale-Location Plot (Bottom-Left)

This plot determines homoscedasticity. The red line is quite flat, with no apparent pattern in spread—though some slight bumps do show. There is no apparent evidence of heteroscedasticity, and the assumption of constant variance is reasonably sustained.

Residuals vs Leverage (Bottom-Right)

This graph uses leverage and Cook's distance to identify influential observations. Observations 8, 10, and 12 have stronger leverage, with point 10 close to a Cook's distance. There may be a few influential spots, but none are significant enough to fundamentally alter the model. Yet these points should be monitored.

The residual diagnostics indicate that the regression model generally meets the assumptions of linear regression. There are no serious violations, however a few observations (most notably 8 and 10) may have a little impact or deviate somewhat from the assumptions.

Statistical Tests

Homoscedasticity test

```
> ncvTest(model_crest)
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.07424735, Df = 1, p = 0.78525
```

Figure 16: Non-constant error variance test.

Inference:

H_0: The residuals have constant variance (homoscedasticity).

H_a: The residuals have non-constant variance (heteroscedasticity).

Test Result

Chi-square = 0.0742

Degrees of freedom = 1

p-value = 0.78525

Since the p-value is much greater than 0.05, we fail to reject the null hypothesis. This indicates that there is no indication of heteroscedasticity, and the assumption of constant variance in residuals is satisfied.

Normality Test

```
        Shapiro-Wilk normality test

data:  model_crest$residuals
W = 0.83777, p-value = 0.01522
```

Figure 17: Shapiro-Wilk Test.

Inference:

Test Result

W = 0.83777

p-value = 0.01522

Since the p-value (0.0152) is less than 0.05, hence we reject the null hypothesis. This shows that the residuals are not normally distributed, which contradicts the normality assumption.

Autocorrelation Test

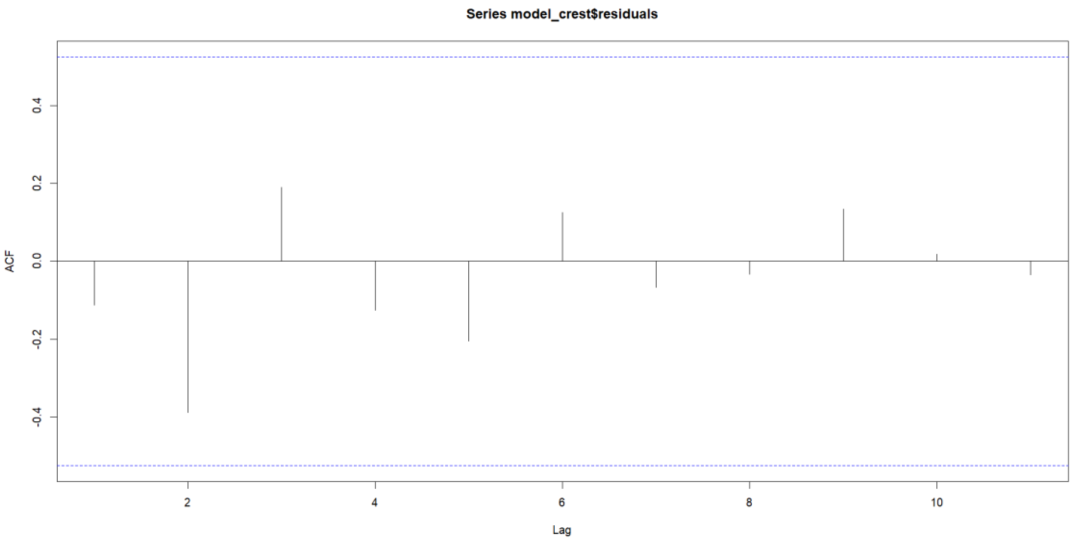

Figure 18: ACF plot

To evaluate the assumption of residual independence, we use the Autocorrelation Function (ACF) plot. In a well-fitted regression model, residuals should be uncorrelated across data.

Inference: Most autocorrelation bars fall within the blue dashed confidence limits (representing the 95% interval), indicating no substantial autocorrelation. Although a few bars (e.g., at lags 2 and 6), indicate minimal spikes, none exceed the significance level, indicating that they are not statistically relevant. There is no apparent pattern or cyclical structure in the residual data. According to the ACF plot, the residuals appear to be randomly distributed and not autocorrelated. As a result, the regression model's argument of independent errors holds true.

```
> durbinWatsonTest(model_crest)
 lag Autocorrelation D-W Statistic p-value
   1      -0.1129573      2.211283    0.87
 Alternative hypothesis: rho ≠ 0
```

Figure 19: Durbin Watson Test.

Hypotheses

H_0: No autocorrelation

H_a : Autocorrelation exists

Inference:

Test Results

Autocorrelation: –0.113

Durbin-Watson statistic (D-W): 2.211

p-value: 0.87

The Durbin-Watson statistic ranges from 0 to 4:

~2 indicates no autocorrelation

< 2 suggests positive autocorrelation

2 suggests negative autocorrelation

The Durbin-Watson test yields a statistic of 2.211 with a p-value of 0.87. We fail to reject the null hypothesis because the p-value is significantly greater than 0.05. This shows that the residuals have no significant autocorrelation, and that the assumption of error independence is satisfied.

Multi-collinearity test

```
          x1        x2        x3
x1 1.0000000 0.7714611 0.9187616
x2 0.7714611 1.0000000 0.6154342
x3 0.9187616 0.6154342 1.0000000
```

Figure 20: Correlation Matrix.

Inference:

- The correlation between x1 (Crest Budget) and x3 (Disposable Income) is 0.92, showing significant multicollinearity.
- x1 and x2 have a strong positive correlation (0.77), while x2 and x3 are moderately correlated (0.62).

The correlation matrix shows evidence of multicollinearity, particularly between x1 and x3. A high correlation between predictors may increase standard errors and make coefficient estimates unstable or difficult to interpret.

```
> vif(model_crest)
       x1        x2        x3
11.421674  2.865841  7.443232
```

Figure 21: VIF values.

Inference:

- The VIF score for x1 (11.42) exceeds the usually accepted threshold of 10, indicating significant multicollinearity, particularly with other predictors such as x3.
- x3 also exhibits a slightly high VIF (7.44), which supports the earlier correlation matrix concern.
- x2 shows no serious concern, with a VIF (2.87) of less than 5.

This multicollinearity may result in unstable or misleading coefficient estimates, so it may be beneficial to explore removing or combining predictors or utilizing model selection processes .

<u>Part E</u>

Model Selection

Forward Selection

```
Start:  AIC=302.64
y ~ 1

       Df  Sum of Sq        RSS    AIC
+ x3    1 2.8411e+10 1.2467e+09 260.27
+ x1    1 2.5611e+10 4.0466e+09 276.75
+ x2    1 1.0637e+10 1.9020e+10 298.42
<none>               2.9658e+10 302.63

Step:  AIC=260.27
y ~ x3

       Df Sum of Sq        RSS    AIC
+ x1    1 171624035 1075039723 260.19
<none>              1246663758 260.27
+ x2    1    574590 1246089168 262.26

Step:  AIC=260.19
y ~ x3 + x1

       Df Sum of Sq        RSS    AIC
+ x2    1 158364757  916674966 259.96
<none>              1075039723 260.19

Step:  AIC=259.96
y ~ x3 + x1 + x2
```

Figure 22: Forward Selection Output.

Inference:

Step 1: Start with Null Model

The process begins with the intercept-only model (y ~ 1). Among the three predictors, x3 (Disposable Income) gives the largest drop in AIC (to 260.27), so it is selected first.

Step 2: Add x1 (Crest Budget)

Adding x1 to the model with x3 further reduces the AIC from 260.27 to 260.19. x1 is added because it provides the next best improvement in model fit.

Step 3: Add x2 (Ratio)

Adding x2 to the current model further reduces the AIC to 259.96 Although the drop is small, it still improves the model slightly.

The forward model selection gives y ~ x3 + x1 + x2.

Using AIC-based forward selection, all three predictors (x3, x1, and x2) were included in the final model. The variable x3 has the greatest influence on reducing AIC and so is the most influential predictor. The improvements from x1 and x2 were less, but they still contributed to improve the model's fit, according to AIC.

Backward Elimination

```
Start:  AIC=259.96
y ~ x1 + x2 + x3

       Df  Sum of Sq        RSS    AIC
<none>                916674966 259.96
- x2    1  158364757 1075039723 260.19
- x1    1  329414202 1246089168 262.26
- x3    1 2109684588 3026359554 274.68
```

Figure 23: Backward Elimination.

Inference:

The process begins with the full model: y ~ x1 + x2 + x3

The AIC of the full model is 259.96.

Removing x2 significantly increases the AIC to 260.19.

Removing x1 increases AIC to 262.26.

Removing x3 significantly raises the AIC to 274.68.

All removals improve AIC; thus, no predictors are eliminated.

Backward elimination retained all three predictors (x1, x2, and x3) in the model because eliminating any variable increased the AIC, suggesting poor model fit. The variable x3 (Disposable Income) is extremely crucial; eliminating it will significantly increase AIC (to 274.68), making it the model's strongest contributor. Thus, the final model is: y ~ x1 + x2 + x3

Stepwise Selection

```
Start:  AIC=302.64
y ~ 1

       Df  Sum of Sq        RSS    AIC
+ x3    1 2.8411e+10 1.2467e+09 260.27
+ x1    1 2.5611e+10 4.0466e+09 276.75
+ x2    1 1.0637e+10 1.9020e+10 298.42
<none>               2.9658e+10 302.63

Step:  AIC=260.27
y ~ x3

       Df  Sum of Sq        RSS    AIC
+ x1    1 1.7162e+08 1.0750e+09 260.19
<none>               1.2467e+09 260.27
+ x2    1 5.7459e+05 1.2461e+09 262.26
- x3    1 2.8411e+10 2.9658e+10 302.63

Step:  AIC=260.19
y ~ x3 + x1

       Df  Sum of Sq        RSS    AIC
+ x2    1  158364757  916674966 259.96
<none>               1075039723 260.19
- x1    1  171624035 1246663758 260.27
- x3    1 2971530267 4046569990 276.75

Step:  AIC=259.96
y ~ x3 + x1 + x2

       Df  Sum of Sq        RSS    AIC
<none>                916674966 259.96
- x2    1  158364757 1075039723 260.19
- x1    1  329414202 1246089168 262.26
- x3    1 2109684588 3026359554 274.68
```

Figure 24: Stepwise selection output.

Inference:

Step 1: Start with Null Model (y ~ 1)

Initial AIC = 302.64

Best variable to add is x3, which drops AIC to 260.27

x3 (Disposable Income) is added.

Step 2: Add x1

Adding x1 to the model (y ~ x3) further reduces AIC to 260.19

This improvement is enough to justify keeping x1.

Step 3: Add x2

Adding x2 reduces the AIC slightly to 259.96, the lowest so far

The model now becomes y ~ x3 + x1 + x2.

Step 4: No More Improvements

Removing any predictor increases the AIC:

Removing x2 → AIC = 260.19

Removing x1 → AIC = 262.26

Removing x3 → AIC = 274.68

Since all removals worsen the model, the process stops.

Final Model after this process is: y ~ x3 + x1 + x2

This is the the same model used by both forward and backward approaches, confirming consistency.

The stepwise selection approach chose the model with all three predictors: x3, x1, and x2, which had the lowest AIC value of 259.96.

x3 (Disposable Income) was the most influential predictor, appearing first and resulting in the biggest decline in AIC.

The inclusion of x1 and x2 resulted in small but steady improvements in model fit, while eliminating any predictor increased the AIC.

Thus, the final model from stepwise selection is the entire model, which confirms its appropriateness among competing alternatives.

Final Model Selected

```
Call:
lm(formula = y ~ x3 + x1 + x2, data = crest)

Residuals:
   Min     1Q Median     3Q    Max 
-24088  -2568   1021   3836  10100 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 34104.559  17654.144   1.932 0.082187 .  
x3             85.926     17.911   4.797 0.000727 ***
x1              3.746      1.976   1.896 0.087243 .  
x2         -30046.343  22859.674  -1.314 0.218066    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9574 on 10 degrees of freedom
Multiple R-squared:  0.9691,	Adjusted R-squared:  0.9598 
F-statistic: 104.5 on 3 and 10 DF,  p-value: 7.537e-08
```

Figure 25: Final Model Summary.

Inference:

Final Regression Equation:

$$\mathbf{y = 34104.56 + 85.93*x3 + 3.75*x1 - 30046.34*x2 + \varepsilon}$$

Here:

x3 = U.S. Personal Disposable Income

x1 = Crest Budget

x2 = Ratio

Crest Budget (x1), Ratio (x2), and Disposable Income (x3) are all included in the final multiple linear regression model, which was selected using forward, backward, and stepwise approaches. The model is statistically significant ($F = 104.5$, $p < 0.001$) and accounts for 97% of the variance in Crest sales ($R^2 = 0.9691$). Among the predictors, x3 (disposable income) is extremely significant ($p = 0.0007$) and has a positive effect on sales. x1 (Crest Budget) is marginally significant ($p = 0.0872$), whereas x2 (Ratio) is non-significant ($p = 0.2181$) but was kept since it contributed to overall model fit. The regression equation demonstrates that more disposable income leads to a significant rise in sales, validating its function as the most useful predictor.

Conclusion

Question 1

In this analysis, a multiple linear regression model was developed to predict the percentage of drug dose retained in a rat's liver based on body weight (X1), liver weight (X2), and dose administered (X3). Initial model fitting revealed that, while the entire model was marginally significant, only X1 and X3 were significant predictors based on individual t-tests. The ANOVA and VIF analyses also revealed that X2 was not a significant contributor, and that multicollinearity existed significantly between X1 and X3. Backward elimination with Fstay = 4 was used to refine the model, resulting in a more compact model that retained X1 and X3 intact. This simplified model increased interpretability while retaining statistical significance and explained a comparable amount of variance. The final model reveals that dose and body weight are significant indicators of liver drug retention, although liver weight can be removed for simplicity and effectiveness.

Question 2

The multiple linear regression analysis of Crest toothpaste sales data found that the final model, which included disposable income (x3), advertising budget (x1), and sales ratio (x2), produced the best fit, as determined by forward, backward, and stepwise selection techniques. Disposable income (x3) was identified as the most significant predictor, with a significant beneficial effect on sales. The advertising budget (x1) was somewhat important, although the sales ratio (x2) was not individually significant but did add to overall model performance. The residual analysis verified the assumptions of linearity, homoscedasticity, and independence, but it also indicated a tiny violation of normality. Multicollinearity was observed, particularly between x1 and x3. Despite this, the resulting model was statistically strong, predicting 97% of the variation in Crest sales, demonstrating the predictive power of income and budget in determining customer sales behaviour.

References

1. https://rmit.instructure.com/courses/140876/files/44499995?wrap=1

2. https://rmit.instructure.com/courses/140876/files/44502029?wrap=1

3. https://rmit.instructure.com/courses/140876/files/44502028?wrap=1

4. https://rmit.instructure.com/courses/140876/files/44500100?wrap=1

5. https://rmit.instructure.com/courses/140876/files/44640414?wrap=1

6. https://rmit.instructure.com/courses/140876/files/44640232?wrap=1

7. https://rmit.instructure.com/courses/140876/files/44640262?wrap=1

8. https://rmit.instructure.com/courses/140876/files/44640310?wrap=1

9. https://rmit.instructure.com/courses/140876/files/44640313?wrap=1

10. https://rmit.instructure.com/courses/140876/files/44640506?wrap=1

Appendix

R Codes Used:

```
# Clear R environment
rm(list = ls())

# importing necessary libraries
library(ggplot2)
library(GGally)
library(car)
library(stats)
library(tidyverse)
library(TSA)
install.packages("rms")
library(rms)

# Question 1
# Load CSV
liver <- read.csv("C:/Users/shamb/OneDrive/Desktop/Regression
Analysis/Assignment 2/liver-1.csv")

# Rename columns
names(liver) <- c("X1", "X2", "X3", "Y")
# Confirming new names
head(liver)

# Part A
# Construct design matrix with intercept
X <- as.matrix(cbind(1, liver$X1, liver$X2, liver$X3))
# Print the design matrix
X

# Taking transpose of X
```

```
x_transpose <- t(X)
x_transpose

# Compute (X'X)^-1
XtX_inverse <- solve( x_transpose %*% X)
XtX_inverse

# Part B
# Fit the full regression model
model <- lm(Y ~ X1 + X2 + X3, data = liver)
# Display model summary
summary(model)

# (Intercept) = 0.2659
# X1 (Body Weight) = -0.0212
# X2 (Liver Weight) = 0.0143
# X3 (Dose Given) = 4.1781

# Part C
# Anova Analysis
anova(model)
# t test
summary(model)

# Part D
# Correlation Matrix of X1,X2,X3
cor(liver[, c("X1", "X2", "X3")])
# Variance Inflation Factor (VIF)
vif(model)

# Part E
# Backward elimination with Fstay = 4
model_back <- step(model, direction = "backward", k = 4)
# View final model
```

```
summary(model_back)


# Question 2
# Crest Toothpaste Sales Data
crest <- data.frame(
  year = 1967:1980,
  y = c(105000, 105000, 121600, 113750, 113750, 128925, 142500, 126000,
162000, 191625, 189000, 210000, 224250, 245000),
  x1 = c(16300, 15800, 16000, 14200, 15000, 14000, 15400, 18250, 17300, 23000,
19300, 23056, 26000, 28000),
  x2 = c(1.25, 1.34, 1.22, 1.00, 1.15, 1.13, 1.05, 1.27, 1.07, 1.17, 1.07, 1.54, 1.59,
1.56),
  x3 = c(547.9, 593.4, 638.9, 695.3, 751.8, 810.3, 914.5, 998.3, 1096.1, 1194.4,
1311.5, 1462.9, 1641.7, 1821.7)
)
# To view first five rows
head(crest)
# Scatter Plot of the data using ggpairs
ggpairs(data = crest[, c("y", "x1", "x2", "x3")],
        title = "Scatterplot Matrix: Crest Sales and Predictors")


# Regression model
model_crest <- lm(y ~ x1 + x2 + x3, data = crest)
summary(model_crest)


# ANOVA of the crest data
anova(model_crest)


# Confidence Intervals (95%)
confint(model_crest)


# Residual Analysis
par(mfrow=c(2,2))
plot(model_crest)
```

```
# Statistical tests
# Non-constant error variance test
ncvTest(model_crest)
# Test for Normally Distributed Errors
shapiro.test(model_crest$residuals)
# Test for Autocorrelated Errors
graphics.off()
acf(model_crest$residuals)
durbinWatsonTest(model_crest)

# Multi-collinearity test
# Correlation matrix of predictors
cor(crest[, c("x1", "x2", "x3")])
# VIF values to detect multi-collinearity
vif(model_crest)

# Model Building: Forward, Backward, Step-wise Selection
# Full model with all predictors
model_full <- lm(y ~ x1 + x2 + x3, data = crest)
# Null model (intercept only)
model_null <- lm(y ~ 1, data = crest)
# Forward Selection
model_forward <- step(model_null,
                      scope = list(lower = model_null, upper = model_full),
                      direction = "forward")
# Backward Elimination
model_backward <- step(model_full, direction = "backward")
# Stepwise Selection (both directions)
model_stepwise <- step(model_null,
                       scope = list(lower = model_null, upper = model_full),
                       direction = "both")
# Choose a final model (e.g., from stepwise)
model_selected <- model_stepwise  # or model_forward / model_backward as appropriate
```

```
# Summary of the selected final model
summary(model_selected)
```